

Subject : Physics
Standard : 11
Total Mark : 100

Shm

Paper Set : 1
Date : 25-09-2026
Time : 0H:0M

Physics - Section A (MCQ)

- (1) A particle is executing *SHM* along a straight line. Its velocities at distances x_1 and x_2 from the mean position are V_1 and V_2 respectively. Its time period is
- (A) $2\pi\sqrt{\frac{x_1^2+x_2^2}{V_1^2+V_2^2}}$
 (B) $2\pi\sqrt{\frac{x_2^2-x_1^2}{V_1^2-V_2^2}}$
 (C) $2\pi\sqrt{\frac{V_1^2+V_2^2}{x_1^2+x_2^2}}$
 (D) $2\pi\sqrt{\frac{V_1^2-V_2^2}{x_1^2-x_2^2}}$
- (2) A particle performs simple harmonic motion with amplitude A . Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from the equilibrium position. The new amplitude of the motion is
- (A) $A\sqrt{3}$ (B) $\frac{7A}{3}$
 (C) $\frac{A}{3}\sqrt{41}$ (D) $3A$
- (3) Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?
- (A) $a + bx - cx^2$ (B) bx^2
 (C) $a - bx + cx^2$ (D) $-bx$
- (4) A particle executes simple harmonic motion with an amplitude of 5 cm . When the particle is at 4 cm from the mean position, the magnitude of its velocity is equal to the magnitude of its acceleration. Then, its periodic time in seconds is
- (A) $\frac{4\pi}{3}$ (B) $\frac{3\pi}{8}$
 (C) $\frac{8\pi}{3}$ (D) $\frac{7\pi}{3}$
- (5) A simple harmonic motion is represented by $y = 5(\sin 3\pi t + \sqrt{3} \cos 3\pi t)\text{ cm}$. The amplitude and time period of the motion are:
- (A) $10\text{ cm}, \frac{2}{3}\text{ s}$ (B) $10\text{ cm}, \frac{3}{2}\text{ s}$
 (C) $5\text{ cm}, \frac{3}{2}\text{ s}$ (D) $5\text{ cm}, \frac{2}{3}\text{ s}$
- (6) A particle performs simple harmonic motion with a period of 2 s . The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}\text{ s}$. The value of a to the nearest integer is:
- (A) 6 (B) 5
 (C) 4 (D) 8
- (7) The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is:
- (A) $\sin(\omega t) + \cos(\omega t)$ (B) $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$
 (C) $\sin^2(\omega t)$ (D) $3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$
- (8) $Y = A \sin(\omega t + \phi_0)$ is the time-displacement equation of a *SHM*. At $t = 0$, the displacement of the particle is $Y = \frac{A}{2}$ and it is moving in the negative direction. Then the initial phase angle ϕ_0 will be
- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
 (C) $\frac{5\pi}{6}$ (D) $\frac{2\pi}{3}$
- (9) A particle executes *S.H.M.* with amplitude a and time period T . The displacement of the particle when its speed is half of its maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is
- (A) 1 (B) 5
 (C) 2 (D) 3
- (10) A particle executes simple harmonic motion represented by the displacement function $x(t) = A \sin(\omega t + \phi)$. If the position and velocity of the particle at $t = 0\text{ s}$ are 2 cm and $2\omega\text{ cm s}^{-1}$ respectively, then its amplitude is $x\sqrt{2}\text{ cm}$, where the value of x is
- (A) 3 (B) 1
 (C) 2 (D) 6
- (11) The displacement of a simple harmonic oscillator after 3 s starting from its mean position is equal to half of its amplitude. The time period of the harmonic motion is ... s.
- (A) 6 (B) 8
 (C) 12 (D) 36
- (12) The equation of a particle executing simple harmonic motion is given by $x = \sin \pi(t + 1/3)\text{ m}$. At $t = 1\text{ s}$, the speed of the particle will be cm s^{-1} (Given: $\pi = 3.14$)
- (A) 0 (B) 157
 (C) 272 (D) 314
- (13) A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6 s . The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is s
- (A) 3 (B) 5
 (C) 1 (D) 2
- (14) When a particle executes simple harmonic motion, the nature of the graph of velocity as a function of displacement will be:
- (A) Circular
 (B) Elliptical
 (C) Sinusoidal
 (D) Straight line
- (15) A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If the time taken by the particle to go from $x = 0$ to $x = A/2$ is 2 s , then the time taken by the particle in going from $x = A/2$ to $x = A$ is s.
- (A) 3 (B) 2
 (C) 1.5 (D) 4
- (16) A particle of mass 250 g executes a simple harmonic motion under a periodic force $F = (-25x)\text{ N}$. The particle attains a

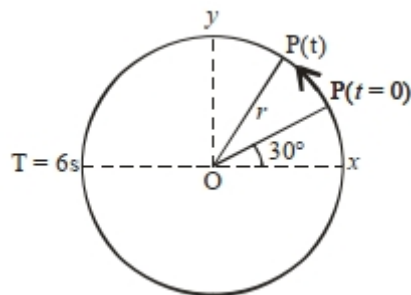
maximum speed of 4 m/s during its oscillation. The amplitude of the motion is cm .

- (A) 20 (B) 40
(C) 60 (D) 80

- (17) The velocity of a particle executing *SHM* varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillations is $\frac{\pi}{7}\text{ s}$. The value of x is (Take $\pi = \frac{22}{7}$)

- (A) 80 (B) 81
(C) 88 (D) 82

- (18) For particle P revolving around the centre O with radius of circular path r and angular velocity ω , as shown in the figure, the projection of OP on the x -axis at time t is



- (A) $x(t) = r \cos(\omega t + \frac{\pi}{6})$ (B) $x(t) = r \cos(\omega t)$
(C) $x(t) = r \sin(\omega t + \frac{\pi}{6})$ (D) $x(t) = r \cos(\omega t - \frac{\pi}{6})$

- (19) A particle executes *S.H.M.* of amplitude A along the x -axis. At $t = 0$, the position of the particle is $x = \frac{A}{2}$ and it moves along the positive x -axis. If the displacement of the particle in time t is $x = A \sin(\omega t + \delta)$, then the value of δ will be:

- (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

- (20) In a linear simple harmonic motion (*SHM*):

- (A) Restoring force is directly proportional to the displacement.
(B) The acceleration and displacement are opposite in direction.
(C) The velocity is maximum at the mean position.
(D) The acceleration is minimum at extreme points.
Choose the correct answer from the options given below:
(A) (A), (B) and (C) only
(B) (C) and (D) only
(C) (A), (B) and (D) only
(D) (A), (C) and (D) only

- (21) A particle executes simple harmonic motion with an amplitude of 4 cm . At the mean position, the velocity of the particle is 10 cm/s . The distance of the particle from the mean position when its speed becomes 5 cm/s is $\sqrt{\alpha}\text{ cm}$, where $\alpha = \dots\dots\dots$

- (A) 11 (B) 22
(C) 12 (D) 15

- (22) A simple harmonic oscillator has an amplitude A and time period $6\pi\text{ s}$. Assuming the oscillation starts from its mean position, the time required by it to travel from $x = A$ to $x = \frac{\sqrt{3}}{2}A$ will be $\frac{\pi}{x}\text{ s}$, where $x = \dots\dots\dots$

- (A) 2 (B) 12
(C) 4 (D) 9

- (23) A particle performs simple harmonic motion with amplitude A . Its speed is increased to three times at an instant when its displacement is $\frac{2A}{3}$. The new amplitude of motion is $\frac{nA}{3}$.

The value of n is

- (A) 7 (B) 8
(C) 9 (D) 10

- (24) The displacement of a particle executing *SHM* is given by $x = 10 \sin(\omega t + \frac{\pi}{3})\text{ m}$. The time period of motion is 3.14 s . The velocity of the particle at $t = 0$ is m/s .

- (A) 10 (B) 15
(C) 20 (D) 25

- (25) A particle is performing simple harmonic motion with an amplitude of 0.06 m and a time period of 3.14 s . The maximum velocity of the particle is cm/s .

- (A) 12 (B) 15
(C) 20 (D) 22

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Shm (Answer Key)

Paper Set : 1
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Physics - Section A (MCQ)

1 - B	2 - B	3 - D	4 - C	5 - A	6 - A	7 - D	8 - C	9 - D	10 - C
11 - D	12 - B	13 - C	14 - B	15 - D	16 - B	17 - C	18 - A	19 - A	20 - A
21 - C	22 - A	23 - A	24 - A	25 - A					

Subject : Physics
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Shm
(Solutions)

Paper Set : 1
Date : 25-09-2026
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Physics - Section A (MCQ)

- (1) A particle is executing SHM along a straight line. Its velocities at distances x_1 and x_2 from the mean position are V_1 and V_2 respectively. Its time period is

- (A) $2\pi\sqrt{\frac{x_1^2+x_2^2}{V_1^2+V_2^2}}$
(B) $2\pi\sqrt{\frac{x_2^2-x_1^2}{V_1^2-V_2^2}}$
 (C) $2\pi\sqrt{\frac{V_1^2+V_2^2}{x_1^2+x_2^2}}$
 (D) $2\pi\sqrt{\frac{V_1^2-V_2^2}{x_1^2-x_2^2}}$

Solution:(Correct Answer:B)

(B) In SHM, the velocity V of a particle at a distance x from the mean position is given by $V = \omega\sqrt{a^2 - x^2}$, where a is the amplitude and ω is the angular frequency. Squaring both sides, we get $V^2 = \omega^2(a^2 - x^2)$.

For distances x_1 and x_2 , we have:

$$V_1^2 = \omega^2(a^2 - x_1^2) \dots (i)$$

$$V_2^2 = \omega^2(a^2 - x_2^2) \dots (ii)$$

Subtracting equation (ii) from (i):

$$V_1^2 - V_2^2 = \omega^2(a^2 - x_1^2 - a^2 + x_2^2)$$

$$V_1^2 - V_2^2 = \omega^2(x_2^2 - x_1^2)$$

$$\omega^2 = \frac{V_1^2 - V_2^2}{x_2^2 - x_1^2}$$

Since the time period $T = \frac{2\pi}{\omega}$, we have $\omega = \frac{2\pi}{T}$.

Substituting this into the equation for ω^2 :

$$\left(\frac{2\pi}{T}\right)^2 = \frac{V_1^2 - V_2^2}{x_2^2 - x_1^2}$$

$$T^2 = 4\pi^2 \left(\frac{x_2^2 - x_1^2}{V_1^2 - V_2^2} \right)$$

$$T = 2\pi\sqrt{\frac{x_2^2 - x_1^2}{V_1^2 - V_2^2}}$$

- (2) A particle performs simple harmonic motion with amplitude A . Its speed is trebled at the instant that it is at a distance $\frac{2A}{3}$ from the equilibrium position. The new amplitude of the motion is

- (A) $A\sqrt{3}$ **(B)** $\frac{7A}{3}$
 (C) $\frac{A}{3}\sqrt{41}$ (D) $3A$

Solution:(Correct Answer:B)

(B) Let the initial amplitude be A and the new amplitude be A' . The velocity v of a particle in simple harmonic motion at a distance x from the equilibrium position is given by $v = \omega\sqrt{A^2 - x^2}$.

At $x = \frac{2A}{3}$, the initial velocity v is $v = \omega\sqrt{A^2 - \left(\frac{2A}{3}\right)^2} =$

$$\omega\sqrt{A^2 - \frac{4A^2}{9}} = \omega\sqrt{\frac{5A^2}{9}} = \frac{\omega A\sqrt{5}}{3}.$$

The new speed is $v' = 3v = \omega A\sqrt{5}$.

Using the formula for the new amplitude A' at the same position $x = \frac{2A}{3}$:

$$(v')^2 = \omega^2(A'^2 - x^2)$$

$$(\omega A\sqrt{5})^2 = \omega^2(A'^2 - \left(\frac{2A}{3}\right)^2)$$

$$5A^2 = A'^2 - \frac{4A^2}{9}$$

$$A'^2 = 5A^2 + \frac{4A^2}{9} = \frac{45A^2 + 4A^2}{9} = \frac{49A^2}{9}$$

$$A' = \sqrt{\frac{49A^2}{9}} = \frac{7A}{3}.$$

- (3) Which of the following expressions corresponds to simple harmonic motion along a straight line, where x is the displacement and a, b, c are positive constants?

- (A) $a + bx - cx^2$ (B) bx^2
 (C) $a - bx + cx^2$ **(D)** $-bx$

Solution:(Correct Answer:D)

(D) In linear S.H.M., the restoring force F acting on a particle must be directly proportional to the displacement x from the equilibrium position and directed towards it. Mathematically, this is expressed as $F = -kx$, where k is a positive force constant.

Comparing this with the given options, the expression $-bx$ (where b is a positive constant) represents the restoring force for simple harmonic motion.

Therefore, the correct option is D.

- (4) A particle executes simple harmonic motion with an amplitude of 5 cm . When the particle is at 4 cm from the mean position, the magnitude of its velocity is equal to the magnitude of its acceleration. Then, its periodic time in seconds is

- (A) $\frac{4\pi}{3}$ (B) $\frac{3\pi}{8}$
(C) $\frac{8\pi}{3}$ (D) $\frac{7\pi}{3}$

Solution:(Correct Answer:C)

(C) Given, amplitude $A = 5\text{ cm}$ and displacement $x = 4\text{ cm}$. The magnitude of velocity in simple harmonic motion is

$$|v| = \omega\sqrt{A^2 - x^2}.$$

The magnitude of acceleration in simple harmonic motion is $|a| = \omega^2 x$.

According to the problem, $|v| = |a|$ at $x = 4\text{ cm}$:

$$\omega\sqrt{A^2 - x^2} = \omega^2 x$$

Substituting the values:

$$\omega\sqrt{5^2 - 4^2} = \omega^2 \times 4$$

$$\omega\sqrt{25 - 16} = 4\omega^2$$

$$\omega \times 3 = 4\omega^2$$

$$\omega = \frac{3}{4}\text{ rad/s}$$

The periodic time T is given by $T = \frac{2\pi}{\omega}$:

$$T = \frac{2\pi}{\frac{3}{4}} = \frac{8\pi}{3}\text{ s}.$$

- (5) A simple harmonic motion is represented by $y = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)\text{ cm}$. The amplitude and time period of the motion are:

- (A)** $10\text{ cm}, \frac{2}{3}\text{ s}$ (B) $10\text{ cm}, \frac{3}{2}\text{ s}$
 (C) $5\text{ cm}, \frac{3}{2}\text{ s}$ (D) $5\text{ cm}, \frac{2}{3}\text{ s}$

Solution:(Correct Answer:A)

(A) Given equation: $y = 5(\sin 3\pi t + \sqrt{3}\cos 3\pi t)\text{ cm}$.

We can rewrite this in the form $y = A\sin(\omega t + \phi)$.

Multiply and divide by 2: $y = 5 \times 2 \left(\frac{1}{2}\sin 3\pi t + \frac{\sqrt{3}}{2}\cos 3\pi t \right).$

Using the identity $\sin(A+B) = \sin A \cos B + \cos A \sin B$, we set $\cos \phi = 1/2$ and $\sin \phi = \sqrt{3}/2$, which gives $\phi = \pi/3$.

So, $y = 10 \sin(3\pi t + \pi/3)$ cm.

The amplitude $A = 10$ cm.

The angular frequency $\omega = 3\pi$ rad/s.

The time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{3\pi} = \frac{2}{3}$ s.

- (6) A particle performs simple harmonic motion with a period of 2 s. The time taken by the particle to cover a displacement equal to half of its amplitude from the mean position is $\frac{1}{a}$ s. The value of a to the nearest integer is:

(A) 6 (B) 5
(C) 4 (D) 8

Solution:(Correct Answer:A)

(A) The equation for displacement in simple harmonic motion starting from the mean position is given by $x(t) = A \sin(\omega t)$, where A is the amplitude and ω is the angular frequency.

Given that the displacement $x = \frac{A}{2}$, we have:

$$\frac{A}{2} = A \sin(\omega t)$$

$$\sin(\omega t) = \frac{1}{2}$$

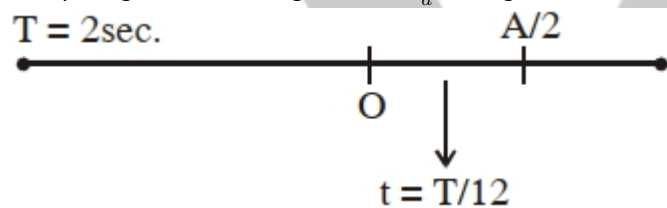
$$\omega t = \frac{\pi}{6}$$

Since $\omega = \frac{2\pi}{T}$, where $T = 2$ s is the time period:

$$\frac{2\pi}{T} \cdot t = \frac{\pi}{6}$$

$$t = \frac{T}{12} = \frac{2}{12} = \frac{1}{6} \text{ s.}$$

Comparing this with the given time $\frac{1}{a}$ s, we get $a = 6$.



- (7) The function of time representing a simple harmonic motion with a period of $\frac{\pi}{\omega}$ is:

(A) $\sin(\omega t) + \cos(\omega t)$ (B) $\cos(\omega t) + \cos(2\omega t) + \cos(3\omega t)$
(C) $\sin^2(\omega t)$ (D) $3 \cos\left(\frac{\pi}{4} - 2\omega t\right)$

Solution:(Correct Answer:D)

(D) The time period of a simple harmonic motion (SHM) is given by $T = \frac{2\pi}{\omega'}$, where ω' is the angular frequency of the SHM.

Given $T = \frac{\pi}{\omega'}$, we have $\frac{\pi}{\omega'} = \frac{2\pi}{\omega'}$, which implies $\omega' = 2\omega$.

We need to find the function that represents SHM with an angular frequency of 2ω .

For option (C): $\sin^2(\omega t) = \frac{1 - \cos(2\omega t)}{2} = \frac{1}{2} - \frac{1}{2} \cos(2\omega t)$. This represents a constant shift plus an SHM with angular frequency 2ω .

For option (D): $3 \cos\left(\frac{\pi}{4} - 2\omega t\right) = 3 \cos\left(2\omega t - \frac{\pi}{4}\right)$. This is a standard SHM equation $x(t) = A \cos(\omega' t + \phi)$ with angular frequency $\omega' = 2\omega$.

Both (C) and (D) have the correct frequency, but (D) is a pure SHM function, whereas (C) includes a constant term.

- (8) $Y = A \sin(\omega t + \phi_0)$ is the time-displacement equation of a SHM. At $t = 0$, the displacement of the particle is $Y = \frac{A}{2}$ and it is moving in the negative direction. Then the initial phase angle ϕ_0 will be

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
(C) $\frac{5\pi}{6}$ (D) $\frac{2\pi}{3}$

Solution:(Correct Answer:C)

(C) The displacement equation is $Y = A \sin(\omega t + \phi_0)$.

At $t = 0$, $Y = A \sin(\phi_0) = \frac{A}{2}$.

This implies $\sin(\phi_0) = \frac{1}{2}$.

Thus, ϕ_0 can be $\frac{\pi}{6}$ or $\frac{5\pi}{6}$.

The velocity of the particle is given by $v = \frac{dY}{dt} = A\omega \cos(\omega t + \phi_0)$.

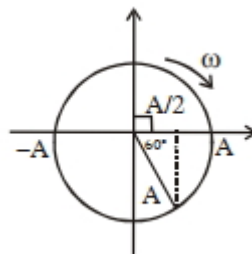
At $t = 0$, $v = A\omega \cos(\phi_0)$.

Since the particle is moving in the negative direction, $v < 0$, which means $\cos(\phi_0) < 0$.

For $\phi_0 = \frac{\pi}{6}$, $\cos(\frac{\pi}{6}) = \frac{\sqrt{3}}{2} > 0$.

For $\phi_0 = \frac{5\pi}{6}$, $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2} < 0$.

Therefore, the initial phase angle is $\phi_0 = \frac{5\pi}{6}$.



- (9) A particle executes S.H.M. with amplitude a and time period T . The displacement of the particle when its speed is half of its maximum speed is $\frac{\sqrt{x}a}{2}$. The value of x is

(A) 1 (B) 5
(C) 2 (D) 3

Solution:(Correct Answer:D)

(D) The velocity of a particle in S.H.M. at displacement y is given by $v = \omega \sqrt{a^2 - y^2}$, where a is the amplitude and ω is the angular frequency.

Maximum speed is $v_{\max} = a\omega$.

Given that the speed $v = \frac{v_{\max}}{2}$, we have $\frac{a\omega}{2} = \omega \sqrt{a^2 - y^2}$.

Squaring both sides: $\frac{a^2}{4} = a^2 - y^2$.

Rearranging for displacement y : $y^2 = a^2 - \frac{a^2}{4} = \frac{3a^2}{4}$.

Thus, $y = \frac{\sqrt{3}a}{2}$.

Comparing this with the given expression $\frac{\sqrt{x}a}{2}$, we find $x = 3$.

- (10) A particle executes simple harmonic motion represented by the displacement function $x(t) = A \sin(\omega t + \phi)$. If the position and velocity of the particle at $t = 0$ s are 2 cm and 2ω cm s⁻¹ respectively, then its amplitude is $x\sqrt{2}$ cm, where the value of x is

(A) 3 (B) 1
(C) 2 (D) 6

Solution:(Correct Answer:C)

(C) The displacement function is $x(t) = A \sin(\omega t + \phi)$.

At $t = 0$, $x(0) = A \sin \phi = 2 \dots (1)$.

The velocity function is $v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \phi)$.

At $t = 0$, $v(0) = A\omega \cos \phi = 2\omega \Rightarrow A \cos \phi = 2 \dots (2)$.

Dividing equation (1) by equation (2):

$$\frac{A \sin \phi}{A \cos \phi} = \frac{2}{2} \Rightarrow \tan \phi = 1 \Rightarrow \phi = 45^\circ$$

Substituting $\phi = 45^\circ$ in equation (1):

$$A \sin 45^\circ = 2 \Rightarrow A \left(\frac{1}{\sqrt{2}}\right) = 2 \Rightarrow A = 2\sqrt{2} \text{ cm.}$$

Comparing $A = 2\sqrt{2}$ cm with $x\sqrt{2}$ cm, we get $x = 2$.

- (11) The displacement of a simple harmonic oscillator after 3 s starting from its mean position is equal to half of its amplitude. The time period of the harmonic motion is ... s.

(A) 6 (B) 8
(C) 12 (D) 36

Solution:(Correct Answer:D)

(D) The displacement equation for a simple harmonic oscillator starting from the mean position is given by $X = A \sin(\omega t)$.

Given that at $t = 3 \text{ s}$, the displacement $X = \frac{A}{2}$.

Substituting these values into the equation: $\frac{A}{2} = A \sin(3\omega)$.

Dividing both sides by A , we get $\sin(3\omega) = \frac{1}{2}$.

Since $\sin(\frac{\pi}{6}) = \frac{1}{2}$, we have $3\omega = \frac{\pi}{6}$.

Thus, $\omega = \frac{\pi}{18}$.

We know that the angular frequency $\omega = \frac{2\pi}{T}$, where T is the time period.

Equating the two expressions for ω : $\frac{2\pi}{T} = \frac{\pi}{18}$.

Solving for T : $T = 2 \times 18 = 36 \text{ s}$.

- (12) The equation of a particle executing simple harmonic motion is given by $x = \sin \pi(t + 1/3) \text{ m}$. At $t = 1 \text{ s}$, the speed of the particle will be cm s^{-1} (Given: $\pi = 3.14$)

(A) 0 (B) 157

(C) 272 (D) 314

Solution:(Correct Answer:B)

(B) The given equation is $x = \sin \pi(t + 1/3) \text{ m}$.

Expanding the equation, we get $x = \sin(\pi t + \pi/3) \text{ m}$.

The velocity v is the derivative of displacement x with respect to time t :

$$v = \frac{dx}{dt} = \frac{d}{dt} [\sin(\pi t + \pi/3)] = \pi \cos(\pi t + \pi/3) \text{ m/s.}$$

At $t = 1 \text{ s}$:

$$v = \pi \cos(\pi(1) + \pi/3) = \pi \cos(4\pi/3) \text{ m/s.}$$

Since $\cos(4\pi/3) = \cos(\pi + \pi/3) = -\cos(\pi/3) = -1/2$:

$$v = \pi \times (-1/2) = -\pi/2 \text{ m/s.}$$

The speed is the magnitude of velocity:

$$|v| = |-\pi/2| = \pi/2 \text{ m/s.}$$

Converting to cm/s ($1 \text{ m} = 100 \text{ cm}$):

$$|v| = (\pi/2) \times 100 = 50\pi \text{ cm/s.}$$

Given $\pi = 3.14$:

$$|v| = 50 \times 3.14 = 157 \text{ cm/s.}$$

- (13) A particle executes simple harmonic motion. Its amplitude is 8 cm and time period is 6 s . The time it will take to travel from its position of maximum displacement to the point corresponding to half of its amplitude, is s

(A) 3 (B) 5

(C) 1 (D) 2

Solution:(Correct Answer:C)

(C) The displacement equation for a particle in SHM starting from the extreme position is $x = A \cos(\omega t)$.

Given amplitude $A = 8 \text{ cm}$ and time period $T = 6 \text{ s}$.

The angular frequency is $\omega = \frac{2\pi}{T} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad/s}$.

We need to find the time t to reach $x = \frac{A}{2} = 4 \text{ cm}$.

Substituting the values: $\frac{A}{2} = A \cos(\omega t) \Rightarrow \cos(\omega t) = \frac{1}{2}$.

This implies $\omega t = \frac{\pi}{3}$.

Substituting $\omega = \frac{\pi}{3}$, we get $\frac{\pi}{3}t = \frac{\pi}{3}$, which gives $t = 1 \text{ s}$.

- (14) When a particle executes simple harmonic motion, the nature of the graph of velocity as a function of displacement will be:

(A) Circular

(B) Elliptical

(C) Sinusoidal

(D) Straight line

Solution:(Correct Answer:B)

(B) For a particle in SHM, its velocity v depends on displacement x as:

$$v = \omega \sqrt{A^2 - x^2}$$

Where ω is the angular frequency and A is the amplitude.

Squaring both sides, we get:

$$v^2 = \omega^2 (A^2 - x^2)$$

$$v^2 = \omega^2 A^2 - \omega^2 x^2$$

Rearranging the terms:

$$v^2 + \omega^2 x^2 = \omega^2 A^2$$

Dividing by $\omega^2 A^2$:

$$\frac{v^2}{(\omega A)^2} + \frac{x^2}{A^2} = 1$$

This equation is of the form $\frac{y^2}{b^2} + \frac{x^2}{a^2} = 1$, which represents an ellipse.

Therefore, the graph between velocity v and displacement x is elliptical.

- (15) A particle executes simple harmonic motion between $x = -A$ and $x = +A$. If the time taken by the particle to go from $x = 0$ to $x = A/2$ is 2 s , then the time taken by the particle in going from $x = A/2$ to $x = A$ is s .

(A) 3 (B) 2

(C) 1.5 (D) 4

Solution:(Correct Answer:D)

(D) The displacement of a particle in simple harmonic motion is given by $x = A \sin(\omega t)$.

For the particle to go from $x = 0$ to $x = A/2$ in time $t_1 = 2 \text{ s}$:

$$A/2 = A \sin(\omega t_1) \Rightarrow \sin(\omega t_1) = 1/2 \Rightarrow \omega t_1 = \pi/6.$$

$$\text{Thus, } \omega(2) = \pi/6 \Rightarrow \omega = \pi/12 \text{ rad/s.}$$

For the particle to go from $x = A/2$ to $x = A$ in time t_2 :

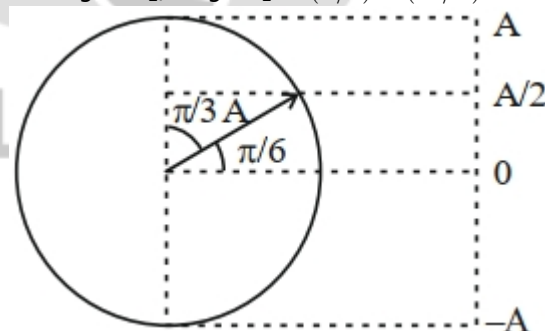
At $x = A/2$, the phase is $\phi_1 = \pi/6$.

At $x = A$, the phase is $\phi_2 = \pi/2$.

The phase difference is $\Delta\phi = \phi_2 - \phi_1 = \pi/2 - \pi/6 = \pi/3$.

Since $\Delta\phi = \omega t_2$, we have $\pi/3 = (\pi/12)t_2$.

Solving for t_2 , we get $t_2 = (\pi/3) \times (12/\pi) = 4 \text{ s}$.



- (16) A particle of mass 250 g executes a simple harmonic motion under a periodic force $F = (-25x) \text{ N}$. The particle attains a maximum speed of 4 m/s during its oscillation. The amplitude of the motion is cm .

(A) 20 (B) 40

(C) 60 (D) 80

Solution:(Correct Answer:B)

(B) Given: mass $m = 250 \text{ g} = 0.25 \text{ kg}$, force $F = -25x$, and maximum speed $v_{\text{max}} = 4 \text{ m/s}$.

According to Newton's second law, $F = ma$, so $ma = -25x$, which gives $a = -\frac{25}{0.25}x = -100x$.

Comparing this with the standard SHM equation $a = -\omega^2 x$, we get $\omega^2 = 100$, so $\omega = 10 \text{ rad/s}$.

The maximum speed in SHM is given by $v_{\text{max}} = \omega A$.

Substituting the values: $4 = 10 \times A$.

Therefore, $A = 0.4 \text{ m}$.

Converting to centimeters: $A = 0.4 \times 100 = 40 \text{ cm}$.

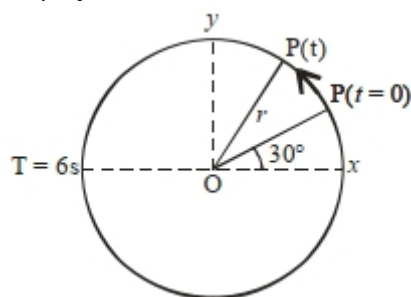
- (17) The velocity of a particle executing SHM varies with displacement (x) as $4v^2 = 50 - x^2$. The time period of oscillations is $\frac{\pi}{7}$ s. The value of x is (Take $\pi = \frac{22}{7}$)

(A) 80 (B) 81
(C) 88 (D) 82

Solution:(Correct Answer:C)

(C) The given equation is $4v^2 = 50 - x^2$.
Dividing by 4, we get $v^2 = \frac{50}{4} - \frac{x^2}{4} = 12.5 - \frac{x^2}{4}$.
Comparing this with the standard SHM velocity equation $v^2 = \omega^2(A^2 - x^2)$, we rewrite our equation as $v^2 = \frac{1}{4}(50 - x^2) = \frac{50}{4}(1 - \frac{x^2}{50})$.
Thus, $\omega^2 = \frac{1}{4}$, which gives $\omega = \frac{1}{2}$ rad/s.
The time period T is given by $T = \frac{2\pi}{\omega} = \frac{2\pi}{1/2} = 4\pi$.
Given $\pi = \frac{22}{7}$, we have $T = 4 \times \frac{22}{7} = \frac{88}{7}$ s.
Comparing this with the given time period $\frac{\pi}{7}$ s, we find $x = 88$.

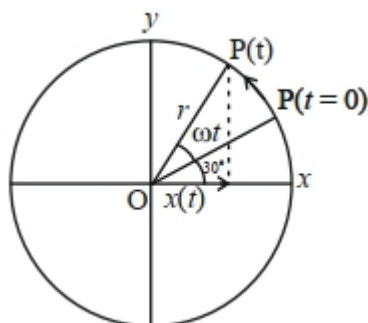
- (18) For particle P revolving around the centre O with radius of circular path r and angular velocity ω , as shown in the figure, the projection of OP on the x -axis at time t is



(A) $x(t) = r \cos(\omega t + \frac{\pi}{6})$ (B) $x(t) = r \cos(\omega t)$
(C) $x(t) = r \sin(\omega t + \frac{\pi}{6})$ (D) $x(t) = r \cos(\omega t - \frac{\pi}{6})$

Solution:(Correct Answer:A)

(A) The particle P is performing uniform circular motion. At $t = 0$, the position vector OP makes an angle of 30° (or $\pi/6$ radians) with the positive x -axis.
At any time t , the particle rotates by an angle ωt in the counter-clockwise direction.
Therefore, the total angle θ made by the position vector OP with the positive x -axis at time t is $\theta = \omega t + 30^\circ = \omega t + \frac{\pi}{6}$.
The projection of the position vector OP on the x -axis is given by $x(t) = r \cos(\theta)$.
Substituting the value of θ , we get $x(t) = r \cos(\omega t + \frac{\pi}{6})$.



- (19) A particle executes S.H.M. of amplitude A along the x -axis. At $t = 0$, the position of the particle is $x = \frac{A}{2}$ and it moves along the positive x -axis. If the displacement of the particle in time t is $x = A \sin(\omega t + \delta)$, then the value of δ will be:

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Solution:(Correct Answer:A)

(A) The displacement equation is given by $x(t) = A \sin(\omega t + \delta)$.

At $t = 0$, the position is $x = \frac{A}{2}$.

Substituting these values: $\frac{A}{2} = A \sin(\omega(0) + \delta) \Rightarrow \sin \delta = \frac{1}{2}$.
This gives two possible values for δ in the range $[0, 2\pi)$: $\delta = \frac{\pi}{6}$ or $\delta = \frac{5\pi}{6}$.

The velocity of the particle is $v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \delta)$.

At $t = 0$, $v(0) = A\omega \cos \delta$.

Since the particle moves along the positive x -axis, the velocity must be positive ($v > 0$), which implies $\cos \delta > 0$.

For $\delta = \frac{\pi}{6}$, $\cos(\frac{\pi}{6}) = \frac{\sqrt{3}}{2} > 0$ (Valid).

For $\delta = \frac{5\pi}{6}$, $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2} < 0$ (Invalid).

Therefore, the correct value is $\delta = \frac{\pi}{6}$.

- (20) In a linear simple harmonic motion (SHM):

(A) Restoring force is directly proportional to the displacement.

(B) The acceleration and displacement are opposite in direction.

(C) The velocity is maximum at the mean position.

(D) The acceleration is minimum at extreme points.

Choose the correct answer from the options given below:

(A) (A), (B) and (C) only

(B) (C) and (D) only

(C) (A), (B) and (D) only

(D) (A), (C) and (D) only

Solution:(Correct Answer:A)

(A) In a linear simple harmonic motion (SHM):

(A) The restoring force is given by $F = -kx$, which means the force is directly proportional to the displacement x and acts in the opposite direction. Thus, statement (A) is true.

(B) The acceleration a is given by $a = -\omega^2 x$. Since the acceleration is proportional to the negative of the displacement, they are always opposite in direction. Thus, statement (B) is true.

(C) The velocity v in SHM is given by $v = \omega \sqrt{A^2 - x^2}$. At the mean position ($x = 0$), the velocity is $v = \omega A$, which is the maximum value. Thus, statement (C) is true.

(D) The magnitude of acceleration is $|a| = \omega^2 |x|$. At the extreme points ($x = \pm A$), the displacement is maximum, so the acceleration is also maximum. Thus, statement (D) is false.

Therefore, statements (A), (B), and (C) are correct.

- (21) A particle executes simple harmonic motion with an amplitude of 4 cm. At the mean position, the velocity of the particle is 10 cm/s. The distance of the particle from the mean position when its speed becomes 5 cm/s is $\sqrt{\alpha}$ cm, where $\alpha = \dots$

(A) 11 (B) 22

(C) 12 (D) 15

Solution:(Correct Answer:C)

(C) The velocity of a particle in simple harmonic motion at the mean position is given by $V_{max} = A\omega$.

Given $A = 4$ cm and $V_{max} = 10$ cm/s, we have $10 = 4\omega$, which gives $\omega = 2.5$ rad/s.

The velocity V at any displacement x from the mean position is given by $V = \omega \sqrt{A^2 - x^2}$.

Substituting the given values: $5 = 2.5 \sqrt{4^2 - x^2}$.

Dividing by 2.5: $2 = \sqrt{16 - x^2}$.

Squaring both sides: $4 = 16 - x^2$.

Therefore, $x^2 = 12$, which means $x = \sqrt{12}$ cm.

Comparing this with $\sqrt{\alpha}$ cm, we get $\alpha = 12$.

- (22) A simple harmonic oscillator has an amplitude A and time period 6π s. Assuming the oscillation starts from its mean position, the time required by it to travel from $x = A$ to $x = \frac{\sqrt{3}}{2}A$ will be $\frac{\pi}{x}$ s, where $x = \dots\dots\dots$

- (A) 2 (B) 12
(C) 4 (D) 9

Solution:(Correct Answer:A)

(A) The equation for displacement starting from the mean position is $x(t) = A \sin(\omega t)$.

Given $T = 6\pi$ s, the angular frequency is $\omega = \frac{2\pi}{T} = \frac{2\pi}{6\pi} = \frac{1}{3}$ rad/s.

At $x = A$, the particle is at the extreme position. The time taken to reach the mean position ($x = 0$) from $x = A$ is $T/4 = (6\pi)/4 = 1.5\pi$ s.

However, the question asks for the time to travel from $x = A$ to $x = \frac{\sqrt{3}}{2}A$.

Using the phasor diagram, the position $x = A \sin(\theta)$ corresponds to the projection on the vertical axis.

At $x = A$, the phase angle $\theta_1 = \frac{\pi}{2}$.

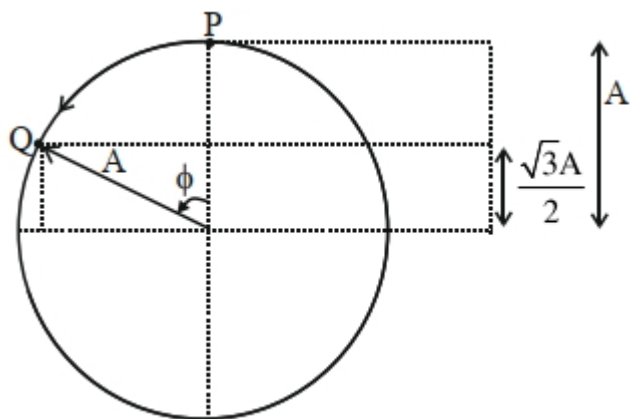
At $x = \frac{\sqrt{3}}{2}A$, $\sin(\theta_2) = \frac{\sqrt{3}}{2}$, so $\theta_2 = \frac{\pi}{3}$.

The change in phase is $\Delta\theta = \theta_1 - \theta_2 = \frac{\pi}{2} - \frac{\pi}{3} = \frac{\pi}{6}$.

Since $\Delta\theta = \omega\Delta t$, we have $\frac{\pi}{6} = \frac{1}{3}\Delta t$.

Therefore, $\Delta t = \frac{3\pi}{6} = \frac{\pi}{2}$ s.

Comparing this with $\frac{\pi}{x}$ s, we get $x = 2$.



- (23) A particle performs simple harmonic motion with amplitude A . Its speed is increased to three times at an instant when its displacement is $\frac{2A}{3}$. The new amplitude of motion is $\frac{nA}{3}$. The value of n is $\dots\dots\dots$

- (A) 7 (B) 8
(C) 9 (D) 10

Solution:(Correct Answer:A)

(A) The velocity of a particle in simple harmonic motion is given by $v = \omega\sqrt{A^2 - x^2}$.

At displacement $x = \frac{2A}{3}$, the velocity is $v =$

$$\omega\sqrt{A^2 - \left(\frac{2A}{3}\right)^2} = \omega\sqrt{A^2 - \frac{4A^2}{9}} = \omega\sqrt{\frac{5A^2}{9}} = \frac{\sqrt{5}A\omega}{3}.$$

When the speed is increased to three times, the new

$$\text{velocity } v' = 3v = 3 \times \frac{\sqrt{5}A\omega}{3} = \sqrt{5}A\omega.$$

Let the new amplitude be A' . The new velocity at the same position $x = \frac{2A}{3}$ is $v' = \omega\sqrt{(A')^2 - x^2}$.

$$\text{Equating the expressions: } \sqrt{5}A\omega = \omega\sqrt{(A')^2 - \left(\frac{2A}{3}\right)^2}.$$

$$\text{Squaring both sides: } 5A^2 = (A')^2 - \frac{4A^2}{9}.$$

$$(A')^2 = 5A^2 + \frac{4A^2}{9} = \frac{45A^2 + 4A^2}{9} = \frac{49A^2}{9}.$$

$$\text{Taking the square root: } A' = \frac{7A}{3}.$$

Comparing this with $\frac{nA}{3}$, we get $n = 7$.

- (24) The displacement of a particle executing SHM is given by $x = 10 \sin\left(\omega t + \frac{\pi}{3}\right)$ m. The time period of motion is 3.14 s.

The velocity of the particle at $t = 0$ is $\dots\dots\dots$ m/s.

- (A) 10 (B) 15
(C) 20 (D) 25

Solution:(Correct Answer:A)

(A) Given, the displacement equation is $x = 10 \sin\left(\omega t + \frac{\pi}{3}\right)$ m.

The time period $T = 3.14$ s. We know that $T = \frac{2\pi}{\omega}$.

Taking $\pi \approx 3.14$, we have $3.14 = \frac{2 \times 3.14}{\omega}$, which gives $\omega = 2$ rad/s.

The velocity v is the derivative of displacement with respect

to time: $v = \frac{dx}{dt} = \frac{d}{dt} \left[10 \sin\left(\omega t + \frac{\pi}{3}\right) \right]$.

$$v = 10\omega \cos\left(\omega t + \frac{\pi}{3}\right).$$

At $t = 0$, the velocity is $v = 10 \times 2 \times \cos\left(0 + \frac{\pi}{3}\right)$.

$$v = 20 \times \cos\left(\frac{\pi}{3}\right) = 20 \times \frac{1}{2} = 10 \text{ m/s}.$$

- (25) A particle is performing simple harmonic motion with an amplitude of 0.06 m and a time period of 3.14 s. The maximum velocity of the particle is $\dots\dots\dots$ cm/s.

- (A) 12 (B) 15
(C) 20 (D) 22

Solution:(Correct Answer:A)

(A) The maximum velocity ($V_{\max}$) of a particle in simple harmonic motion is given by the formula: $V_{\max} = \omega A$.

Here, ω is the angular frequency and A is the amplitude.

We know that $\omega = \frac{2\pi}{T}$, where T is the time period.

Given: $A = 0.06$ m and $T = 3.14$ s $\approx \pi$ s.

Substituting the values:

$$V_{\max} = \left(\frac{2\pi}{\pi}\right) \times 0.06 = 2 \times 0.06 = 0.12 \text{ m/s}.$$

To convert the velocity into cm/s, we multiply by 100:

$$V_{\max} = 0.12 \times 100 = 12 \text{ cm/s}.$$